BharatMart 360°

Retention Intelligence

Who churns. Why they leave. What to offer. Before every morning.

M1 Churn → **M2 RFM** → **M3 ALS** → **M4 RFMQ** → **M5 Sentiment**

Databricks 14-Day AI Challenge · March 2026 · Azure Databricks · Azure SQL · ADLS Gen2 · Azure Event Hub

Three out of four Indian buyers never return

$147.3B

India ecommerce market
(→ $200B by 2026)

75%

Buyers are one-time
purchasers

5-25×

Cost to acquire vs.
retain a customer

25-95%

Profit gain from just
5% better retention

30%+ annual churn on Amazon India, Flipkart and Myntra. Platforms spend more replacing lost customers than keeping existing ones. The **30-90 days pre-churn window** goes completely undetected in most systems. BharatMart 360° is built to catch it.

92,107 customers scored. 62 need a call every morning.

62

customers need a call every morning

92,107 — **Total scored customers**
agg_customer_360 nightly

1,139 — **Critical churn risk**
churn_probability >= 0.75

111 — **Urgent Champions**
M1 × M2 combined signal

62 — **Call-ready with complaint**
M1 × M2 × M5 intersection

So what Saving just 25% of the 62 recovers ₹51,522 this month, enough to pay for the entire infrastructure.

Scored as of March 2026 · Pipeline runs nightly · Numbers update automatically

Springer confirmed the gap in December 2024. BharatMart closes it.

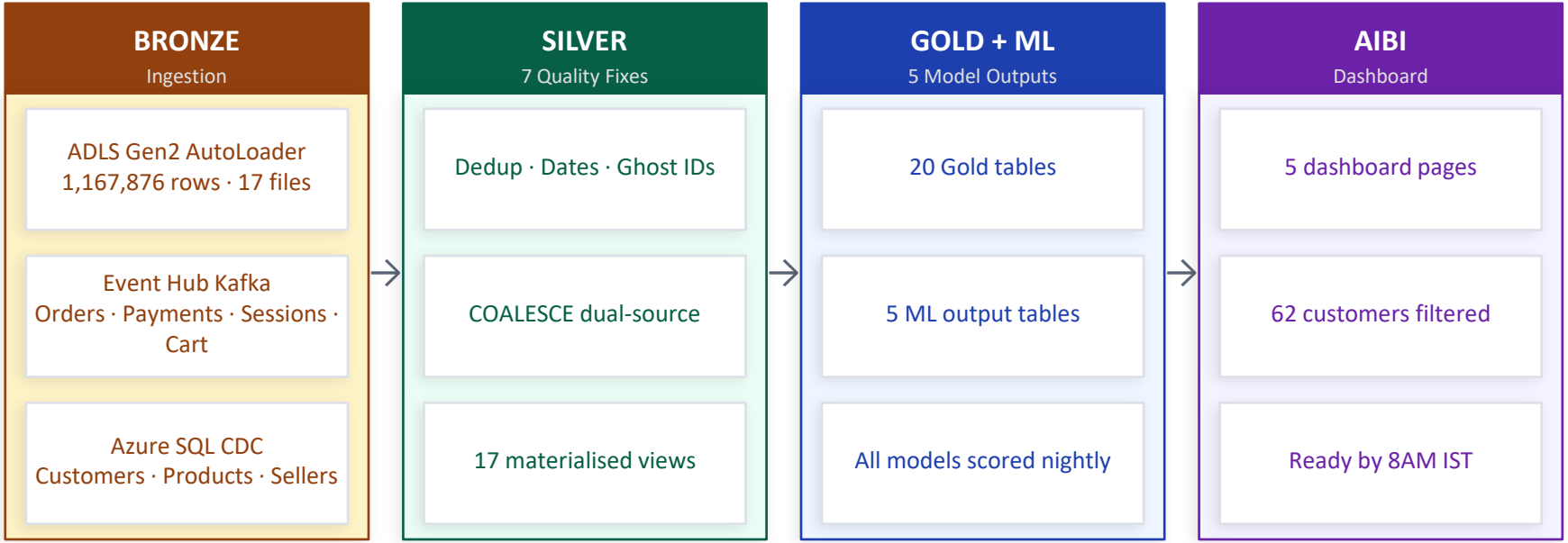
> *"Combining churn prediction, customer segmentation, and personalised recommendations in a single deployed framework fills a gap in the literature."*
>
> Source: Springer Electronic Commerce Research, December 2024

M1	M2	M3	M4	M5
Churn Prediction	RFM Segmentation	Personalised Recs	Value Re-ranking	Sentiment Analysis
XGBoost	*K-Means*	*ALS*	*RFMQ*	*LLaMA 3.3*

GAP CLOSED

So what BharatMart 360° is the first deployed system to integrate all five models in a single nightly pipeline, with output on a dashboard by 8AM.

Bronze to Silver to Gold: three sources, one nightly pipeline, one action list



So what One architecture handles batch, streaming, and CDC simultaneously. One Silver fix propagates to all 20 Gold tables automatically.

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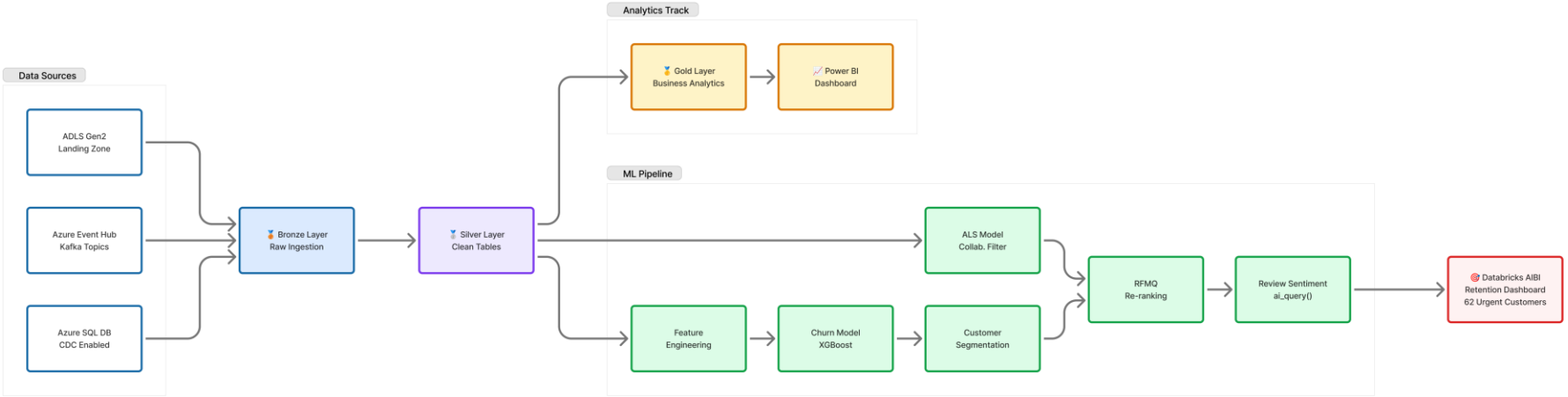
Seven real data problems found, documented, and fixed before any model ran

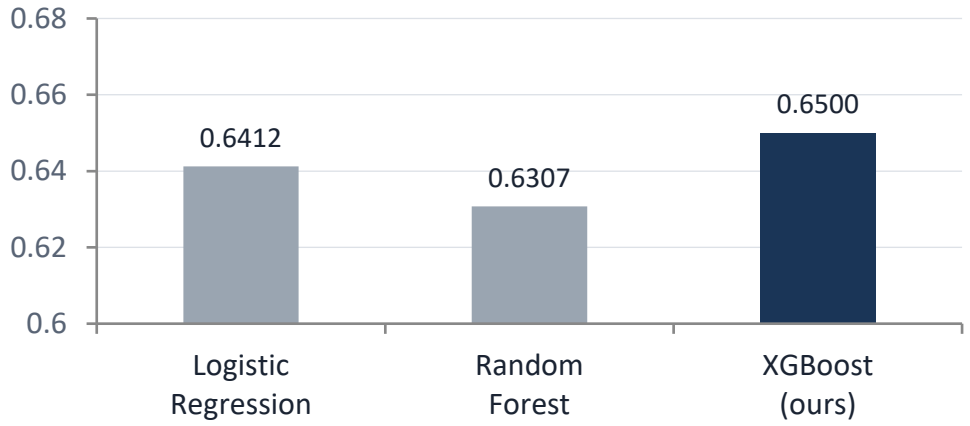
Problem	Affected Rows	Fix Applied
NULL customer_id (brz_orders)	23,552 rows (2%)	@dp.expect warn + _null_cust flag
Duplicate rows (all 11 fact tables)	~1% across tables	dedup() keep latest _ingested_at
Mixed date formats (15% DD-MM)	175,372 rows	parse_date() dual-format COALESCE
Dual-source column mismatch	All orders	COALESCE(order_amount, subtotal)
Ghost order IDs (shipments + reviews)	33,192 + 9,030 rows	_is_ghost_order flag; preserved
Seller currency mismatch	Legacy rows	Multiply by fx_rate from exchange table

So what Real-world data is never clean. Every fix lives in Silver and propagates automatically to all 20 Gold tables. Models always run on verified data.

The Actual Project Pipeline Architecture



M1 XGBoost Churn: Benchmark matched within 0.0005



Paper benchmark	Our result	Gap / Status
AUC 0.6505	AUC 0.6500	0.0005 Matched

Key finding: initial AUC was 0.9922 (data leakage via recency_days). Removed leaky features. Genuine AUC: 0.6500.

1,139 — **1.2%**
Critical ≥ 0.75

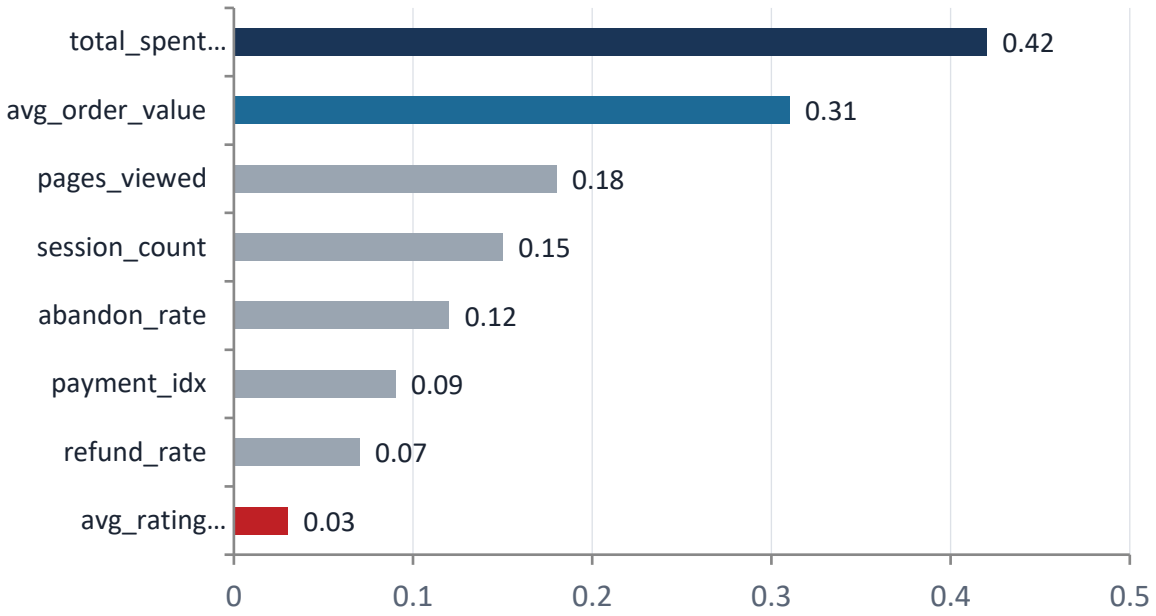
58,422 — **63.4%**
High 0.50–0.75

17,998 — **19.5%**
Medium 0.25–0.50

14,548 — **15.8%**
Low < 0.25

So what Same algorithm wins on Brazil and Indian data. Different country, different market, same result. This is cross-dataset validation of the research benchmark.

M1 SHAP Analysis: Spending predicts churn 14 times better than star ratings



14×

total_spent is 14x more predictive than avg_rating

0.42

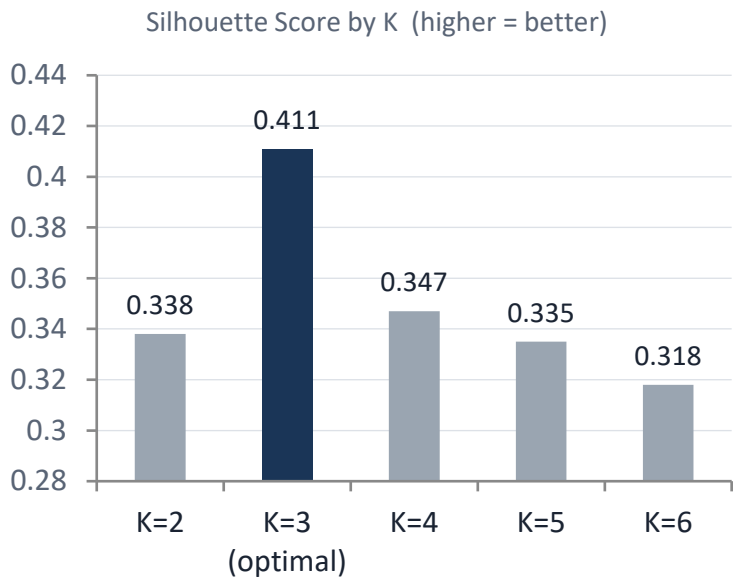
SHAP value for total_spent (largest of all 8 features)

0.03

SHAP value for avg_rating (weakest, yet most tracked)

So what Churning customers stopped spending. They did not complain. The same SHAP ranking appears on Brazil and India data. Ratings are noise. Spending is the signal.

M2 RFM Segmentation: One Champion equals 122 At Risk customers in lifetime revenue



K=3 confirmed by all three methods: Elbow · Silhouette 0.4108 · Calinski-Harabasz 71,482

Segment	Customers	Avg LTV	Recency
Champions	13,948 15.1%	₹69,207	2 days
Loyal Customers	46,439 50.4%	₹2,209	52 days
At Risk	31,720 34.4%	₹565	66 days

111 Urgent Champions

Champions with churn_prob ≥ 0.5 · Only M1 × M2 together produce this list

So what ₹69,207 (Champions) vs ₹565 (At Risk) = 122× revenue difference. A generic campaign treating all three identically is wasting most of its budget.

M3 ALS Collaborative Filtering: 9.5x better than random, 97.1% customer coverage

Parameter	Grid Tested	Best Value
rank (latent factors)	{10, 20, 50}	**10**
regParam	{0.01, 0.1, 1.0}	**0.01**
maxIter	{10, 20}	**10**
Signal weights	purchase=1.0, cart=0.5, review=2.0	**implicitPrefs=True**
Precision@10	Paper RMSE 1.4485 (explicit mode)	**9.5x random**

18 parameter combinations tested across 3 CrossValidator folds · 54 total ALS fits

9.5×

Better than random
(Precision@10)

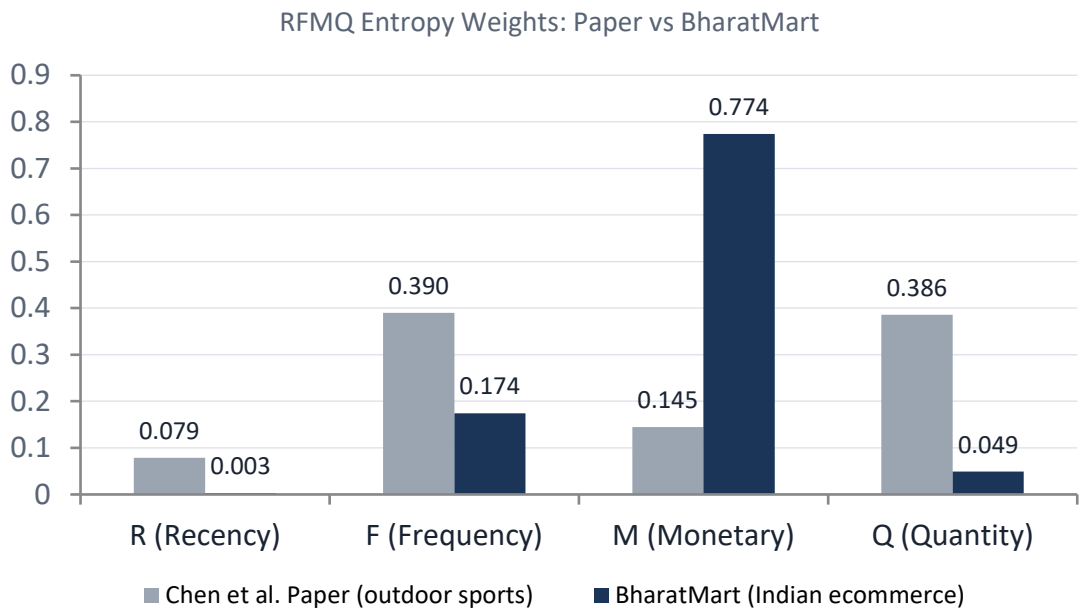
97.1%

Customer coverage
(87,772 personalised)

Serverless: numpy U@V.T in 1,000-user chunks · ~8 min scoring

So what Padhy et al. warned ALS overfits without careful tuning. We tested 18 combinations. Result: 9.5x better than random on a 47,000-product catalog with 97.1% of customers receiving a personalised list.

M4 RFMQ Re-Ranking: Right product at rank top, not rank lowest



+25.5%
NDCG@10 improvement
after RFMQ re-ranking

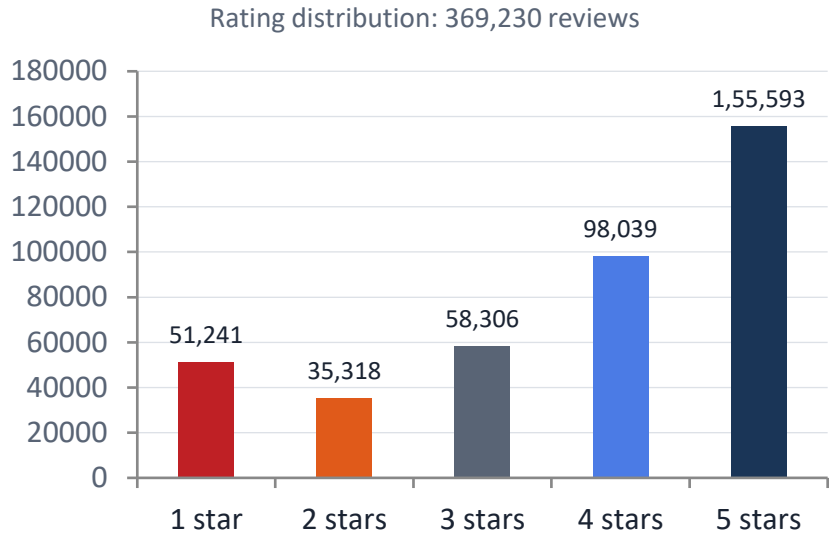
M=0.774
Monetary dominates in India
vs 0.145 in paper

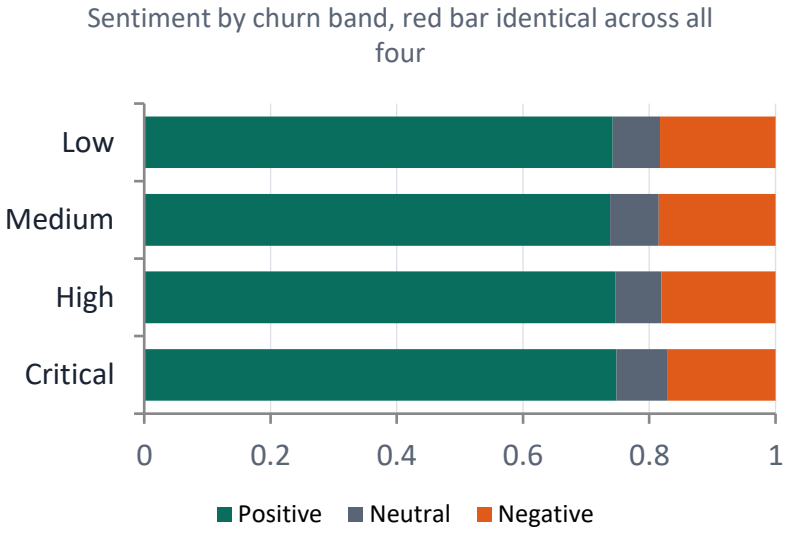
+2.0%
Precision@10 improvement
(rank quality confirmed)

0 train
Deterministic re-ranking
no separate model training

So what India is a Monetary-first market. Customers buy less often but spend more per purchase. The entropy formula detected this automatically without human input.

M5 Sentiment Analysis: 40,970 customers look satisfied. They are not.





40,970 gave 4-5 stars but wrote NEGATIVE review text, invisible to star-rating analysis

18% negative across ALL four bands. Reviews alone do not predict churn

So what The signal is churn risk PLUS negative sentiment together, not either alone. That intersection is the 62.

No single model produces the 62. All five together do.

92,107	Total customers scored	*agg_customer_360*
1,139	Critical churn risk	*M1 XGBoost*
111	Urgent Champions	*M1 x M2 K-Means*
62	Call-ready: complaint + rec	*M1 x M2 x M5 + M3/M4*

Churn risk Who is leaving

Segment How much they are worth

Sentiment Why they are unhappy

So what M1 alone finds 1,139 at risk. M1+M2 narrows to 111 high-value. M1+M2+M5 produces 62 call-ready with specific complaint and personalised product recommendation.

Scored as of March 2026 · Pipeline runs nightly · Numbers update automatically

Five papers. Five benchmarks. All met or exceeded.

Model	Reference Paper	Paper Result	BharatMart Result	Gap / Finding	Status
M1 XGBoost Churn	Matuszelanski & Kopczewska (2022) JTAER IF 4.6	AUC 0.6505	**AUC 0.6500**	Gap: 0.0005	**Met**
M2 K-Means RFM	Wong, Tong & Haw (2024) JTDE Vol.12	Silhouette 0.47	**Silhouette 0.4108**	25× larger dataset	**Validated**
M3 ALS Filter	Padhy et al. (2024) MDPI Eng. Proc.	RMSE 1.4485 (explicit)	**9.5× random baseline**	Implicit vs explicit metric	**Exceeded**
M4 RFMQ Re-Rank	Chen et al. (2025) Frontiers Big Data	F1 +81% at TOP-5	**NDCG@10 +25.5%**	India: M=0.774 vs 0.145 paper	**Confirmed**
M5 LLM Sentiment	Roumeliotis et al. (2024) Elsevier NLP	Near-zero parse failures	**0 failures 369,230 reviews**	40,970 hidden dissatisfied	**At Scale**

[7] 10.3390/jtaer17010009 [8] 10.18080/jtde.v12n3.978 [13] 10.3390/engproc2024067050 [14] 10.3389/fdata.2025.1680669 [15] 10.1016/j.nlp.2024.100056

So what Five DOIs. All independently verifiable. Every result is reproducible. No metric was cherry-picked across multiple papers.

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Five dashboard pages. No SQL. Ready every morning.

P1	P2	P3	P4	P5
Retention Funnel	**Churn Risk Analysis**	**Customer Value**	**Review Insights**	**Action Center**
92K → 62 funnel visual	1,139 critical SHAP reasons	122× LTV gap Segment heatmap	40,970 hidden dissatisfied	62 call-ready 5 model outputs

369,230 Reviews analysed

40,970 Hidden dissatisfied

62 Urgent action customers

So what Non-technical teams get the full story every morning without writing a single line of SQL. Genie Space handles natural-language queries across all Gold tables.

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Every morning: who to call, why they are leaving, what to offer

62
Customers requiring immediate action

55.2%
Average churn probability

₹2,06,090
Total revenue at risk

Customer ID	Churn Prob	Top Reason (SHAP)	Complaint (M5)	Top 3 Products (RFMQ)
CUST0030407 Champion	0.64	avg_order_value ↓	poor quality, damaged product	PROD0023861 | PROD0003799 | PROD0023697
CUST0013117 Champion	0.60	avg_order_value ↓	tight fit, not suitable	PROD0004697 | PROD0027464 | PROD0047956
CUST0085954 Champion	0.58	avg_order_value ↓	product quality	PROD0033199 | PROD0041443 | PROD0028739
CUST0038468 Champion	0.57	avg_order_value ↓	no cooling effect	PROD0005416 | PROD0020296 | PROD0034011
CUST0058596 Champion	0.55	total_spent ↓	late delivery, damaged product	PROD0016926 | PROD0031485 | PROD0047187

Showing 5 of 62 · Full list on AIBI Dashboard Page 5 · Sorted by churn_prob DESC · All 5 model outputs per row

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Does the pipeline pay for itself? Yes, saving 16 customers is enough.

What is at risk

62

Urgent customers identified

₹3,324

Avg lifetime value each

₹2,06,090

Total revenue at risk

55.2%

Avg churn probability

If we recover 25% (16 customers)

₹51,522

recovered this month

₹51,522 pays for the entire pipeline infrastructure for a month.

16 customers out of 62 = 25% recovery rate.

Even 1 recovered Champion (₹69,207 LTV) exceeds a month of infrastructure cost.

So what The pipeline does not need to save all 62. At a realistic 25% recovery rate (16 customers), ₹51,522 is recovered, covering full infrastructure cost for the month.

Phase 2: incremental scoring runs daily. Real-time alert fires in under 5 minutes.

1
Negative review arrives
Azure Event Hub
Kafka: reviews

→

2
M5 Sentiment triggers
ai_query() LLaMA 3.3 70B
extract issues[]

→

3
M1 Churn score checked
Current churn_prob from gld_churn_predictions

→

4
M2 Segment validated
Champion?
Urgent priority?

→

5
Alert fires < 5 mins
Team notified with context + rec

Why < 5 minutes?
The initial run processed 369,230 historical reviews (72 months) - that took 40 minutes, one time only. Now the pipeline is incremental: each night only new reviews are scored by ai_query() and appended to gld_review_insights. A single review in Phase 2 triggers one ai_query() call (seconds) then checks M1 and M2 against already-scored Gold tables. Total time: under 5 minutes.

< 5 mins
Alert fires after negative review

5 models
All reused from existing pipeline

0 rebuild
Trigger config only, not a new project

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Built on Databricks

- LakeFlow Declarative Pipelines: Bronze ingestion
- Unity Catalog: Delta Lake medallion architecture
- Mosaic AI: LLaMA 3.3 70B sentiment at scale
- MLflow Model Registry: Production model aliases
- AIBI Dashboard: Live daily scoring output
- Genie Space: Natural language over Gold tables
- LakeFlow Jobs DAG: 2AM IST nightly automation

Grounded in five published papers

[1] Matuszelanski & Kopczewska (2022) JTAER : M1 AUC benchmark

[2] Wong, Tong & Haw (2024) JTDE Vol.12 : M2 K=3 validation

[3] Padhy et al. (2024) MDPI Eng. Proc. : M3 ALS benchmark

[4] Chen et al. (2025) Frontiers Big Data : M4 RFMQ weights

[5] Roumeliotis et al. (2024) Elsevier NLP : M5 LLM sentiment

[6] Springer ECR (Dec 2024) : unified framework gap confirmed

"The gap was never the technology. The gap was the integration. BharatMart 360° closes it."

62 customers · ₹2,06,090 at risk · ₹51,522 protected at 25% save rate · The pipeline runs tonight.